SHOPPING CART SYSTEM

PROJECT REPORT

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Bachelor of Technology

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CERTIFICATE

This is to certify that the project entitled **"SHOPPING CART SYSTEM"** is a bonafide work carried out by **Kanakam Pavani**, Roll Number **160125771094**, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Data Science during the academic year 2025-26.

Project Guide	**Head of Department**	**Principal**
(Signature)	(Signature)	(Signature)

ACKNOWLEDGEMENT

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1. ABSTRACT

The Shopping Cart System is a web-based application developed using HTML, CSS, and JavaScript to facilitate online shopping operations. The application allows users to browse products, add items to a shopping cart, update product quantities, remove products, and view the total cost of selected items.

The system provides a simple and user-friendly interface that helps customers manage products before checkout. The project demonstrates the practical implementation of front-end web technologies in developing an e-commerce application and serves as a foundation for more advanced online shopping systems.

2. INTRODUCTION

E-commerce has become an important part of modern business and daily life. Online shopping platforms allow customers to purchase products conveniently from anywhere at any time.

A shopping cart is one of the essential components of an online shopping system. It enables customers to collect products, review their selections, and calculate the total amount before placing an order.

The Shopping Cart System provides a simple digital solution for managing product purchases. This project demonstrates how HTML, CSS, and JavaScript can be used to build an interactive and responsive shopping platform.

3. OBJECTIVES

The main objectives of the project are:

- To develop a simple shopping cart application.
- To display available products to users.
- To allow users to add products to the cart.
- To manage product quantities dynamically.
- To calculate the total purchase amount automatically.
- To remove unwanted products from the cart.
- To improve understanding of front-end web development.
- To provide an efficient shopping experience.

4. SCOPE OF THE PROJECT

The Shopping Cart System is suitable for small-scale online shopping applications and educational purposes.

Application Areas

- Online Retail Stores
- E-commerce Websites
- College Mini Projects
- Business Product Catalogs
- Shopping Demonstration Applications

The current version operates within a single browser environment and focuses on shopping cart management rather than complete commercial transactions.

5. TECHNOLOGIES USED

5.1 HTML

HTML (HyperText Markup Language) is used to create the structure of the web application. It defines product cards, shopping cart sections, buttons, and display areas.

5.2 CSS

CSS (Cascading Style Sheets) is used for designing the user interface. It manages layouts, fonts, colors, spacing, and responsiveness.

5.3 JavaScript

JavaScript provides dynamic functionality for adding products, updating quantities, removing items, and calculating totals.

5.4 Visual Studio Code

Visual Studio Code is used as the development environment for writing and editing the source code.

5.5 Web Browser

A modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox is used to run and display the system.

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Desktop Computer or Laptop
- Keyboard and Mouse
- Minimum 4 GB RAM
- Basic Storage Space

Software Requirements

- Windows / Linux / macOS
- Visual Studio Code
- Google Chrome or Microsoft Edge
- Live Server Extension (Optional)

No database or backend server is required for the basic implementation.

7. SYSTEM DESIGN

The project consists of three main files:

index.html

Defines the webpage structure and shopping interface.

style.css
Controls layouts, colors, spacing, and overall appearance.

script.js
Contains the shopping cart logic and user interaction functionality.

SYSTEM WORKFLOW
1. User opens the Shopping Cart System.
2. Products are displayed on the homepage.
3. User selects a product.
4. User clicks the “Add to Cart” button.
5. Product is added to the shopping cart.
6. User can increase or decrease quantity.
7. Total amount updates automatically.
8. User may remove products from the cart.
9. User proceeds to checkout.

8. MODULES OF THE PROJECT

8.1 Product Display Module
Displays product information including product name, image, description, and price.

8.2 Add to Cart Module
Allows customers to add selected products to the shopping cart.

8.3 Cart Management Module
Maintains the list of products selected by the customer and updates quantity values.

8.4 Quantity Update Module
Enables users to increase or decrease product quantities according to their requirements.

8.5 Price Calculation Module
Calculates subtotal and total cost based on product prices and quantities.

8.6 Remove Product Module
Allows customers to remove unwanted products from the shopping cart.

8.7 Checkout Module
Displays the final order summary before purchase completion.

9. IMPLEMENTATION

The application is implemented using HTML, CSS, and JavaScript.

HTML creates the structure and components of the application. CSS enhances the appearance and responsiveness of the interface. JavaScript handles user interactions and shopping cart operations.

The main functions include:
• addToCart()
• removeItem()
• updateQuantity()
• calculateTotal()
• displayCart()

These functions work together to provide a seamless shopping experience.

10. EXECUTION PROCEDURE

The application can be executed by following these steps:

1. Install Visual Studio Code.
2. Create a folder named Shopping-Cart-System.
3. Create the following files: index.html, style.css, script.js.
4. Insert the required code into each file.
5. Save all files.
6. Open the project using a web browser or Live Server.
7. Test adding, removing, and updating products.

The application executes directly in the browser without requiring compilation.

11. TESTING

Testing is performed manually through a web browser.

12. ADVANTAGES

• Simple and user-friendly interface.
• Easy to implement and maintain.
• Automatic price calculation.
• Reduces manual billing effort.
• Provides quick shopping management.
• No database required for the basic version.
• Useful for learning front-end development.
• Easily expandable with additional features.

13. LIMITATIONS

• No database integration.
• No user authentication.
• No online payment support.

• Data is not stored permanently.
• Only single-user support.
• Limited functionality compared to commercial e-commerce systems.

14. FUTURE ENHANCEMENTS

The project can be enhanced by adding:

• User Login and Registration
• Product Search Functionality
• Shopping Wishlist
• Product Categories and Filters
• Database Integration
• Online Payment Gateway
• Order History Tracking
• Product Reviews and Ratings
• Admin Dashboard
• Multi-user Support
• Inventory Management
• Email Notifications

15. CONCLUSION

The Shopping Cart System is a simple web-based application developed using HTML, CSS, and JavaScript to manage online purchases effectively. The system allows users to browse products, add items to a cart, update quantities, remove products, and calculate the total purchase cost automatically.

The project demonstrates the practical use of front-end web development technologies in creating an interactive e-commerce solution. Although the current version has certain limitations, it provides a strong foundation for future enhancements such as user authentication, database connectivity, payment processing, and advanced shopping features.

Testing – Test Cases

Test Case	Expected Result
Add Product	Product is added to the cart
Add Multiple Products	All selected products appear in the cart
Increase Quantity	Product quantity increases correctly
Decrease Quantity	Product quantity decreases correctly
Remove Product	Product is removed from the cart
View Total Amount	Correct total amount is displayed
Empty Cart	Cart shows no products

Test Case	Expected Result
Add Same Product Again	Quantity increases automatically

SOURCES / PROJECT SCREENSHOT

Project source: HTML, CSS, and JavaScript implementation developed for the Shopping Cart System. A screenshot of the executed application can be inserted below.

[SCREENSHOT TO BE INSERTED HERE]